

AI-Assisted Data Analytics Final Project

Project Overview:

- End-to-end AI-assisted data analytics workflow
- Starts from raw data and ends with actionable insights
- Combines data cleaning, visualization, predictive modeling, and AI
- Results presented through an interactive dashboard and face-cam walkthrough

Dataset Selection:

- Structured tabular dataset (Sales, HR, Finance, or Training Outcomes)
- Includes numerical and categorical variables
- It contains missing values and outliers
- Has a clearly defined target variable for prediction

Data Cleaning & Preparation (AI-Assisted):

- Handling missing values and duplicates
- Correcting data types and formatting
- Detecting and treating outliers using statistical methods
- Feature engineering to enhance analysis
- AI-assisted recommendations for cleaning and anomaly detection

Exploratory Data Analysis & Visualization

- Analysis of data distributions and trends
- Identification of relationships between variables
- Use of bar charts, line charts, histograms, and scatter plots
- Extraction of key insights and performance patterns

Predictive Modeling:

- Prediction of a target variable
- Implementation of Logistic Regression, Decision Tree, and Random Forest, Linear Regression, SVM models
- Evaluation using accuracy, and confusion matrix

AI-Generated Insights & Summary:

- Automatic generation of natural-language insights
- Explanation of trends and predictive results
- Business-oriented recommendations for decision making

Interactive Dashboard:

- Developed using Power BI or Tableau
- Displays KPIs and summary metrics
- Includes filters and slicers for dynamic exploration
- Supports drill-down analysis and predictive insights

Face-Cam Walkthrough Presentation:

- Video explanation of the complete project workflow
- Covers objectives, cleaning, analysis, and modeling
- Live demonstration of the interactive dashboard
- Explanation of AI-generated insights

Tools & Technologies:

- Data Cleaning: Python (Pandas, NumPy), Excel
- Visualization: Power BI or Tableau
- Modeling: Scikit-learn
- AI Assistance: Generative AI
- Presentation: Screen recording with face-cam

Final Deliverables:

- Cleaned and prepared dataset
- Exploratory analysis and visualizations
- Predictive model with evaluation metrics
- Interactive dashboard
- AI-generated insights and summary
- Face-cam walkthrough video

Project Value

- Demonstrates full data analytics lifecycle
- Shows practical integration of AI in analytics
- Highlights strong data storytelling skills